

JEE MAIN REVISION MASTER: APPLICATION OF DERIVATIVES 2026 2 QUESTIONS 3.33% WEIGHTAGE ↑99.4%

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 33 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Kcet Application Of Derivatives - Core Application	33 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
-----------------	--

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Kcet Application Of Derivatives - Core Application (Very High Yield — 33 PYQs)

Core Principle: Key problem pattern in Kcet Application Of Derivatives.

■ ■ Caution / JEE Pitfall: Watch for algebraic and sign errors.

■ Speed Trick: Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "A YouTube short video is getting viral according to $f(t)=-2t^3+3t^2+5$. At what time does the video get maximum number of shares? (\$t...\$ " [Ans: N/A]
- [2026 Shift] Q#2 (MCQ): "In a Mahakumbh, a drone camera is moving along $3y=x^3-3$. When y -coordinate changes 9 times as fast as x -coordinate, it captures..." [Ans: C]
- [2025 Shift] Q#3 (MCQ): "The function $f(x)=\tan x-x...$ " [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

A YouTube short video is getting viral according to $f(t) = -2t^3 + 3t^2 + 5$. At what time does the video get maximum number of shares? (t is in hours)

- (A) 1
- (B) 2
- (C) 3
- (D) 4

Correct Answer: None

Solution Analysis: The number of shares is given by $f(t) = -2t^3 + 3t^2 + 5$. To find when the video gets the maximum number of shares, we differentiate $f(t)$ with respect to t . $f'(t) = \frac{d}{dt}(-2t^3 + 3t^2 + 5) = -6t^2 + 6t$. For maximum or minimum value, put $f'(t) = 0$: $-6t^2 + 6t = 0$, $6t(1-t) = 0$. So, $t = 0$ or $t = 1$. Now check the nature using the second derivative. $f''(t) = -12t + 6$. At $t = 1$, $f''(1) = -12(1) + 6 = -6$. Since this is negative, $f(t)$ has a maximum at $t = 1$. So, the video gets the maximum number of shares after 1 hour. Correct option: Option A.